

Foundational Research & Advances in Quantum Phase Estimation (QPE)

Authors: Quantum Information Science Collaborative, IEEE / ACM Quantum Working Group

Published in: ACM Transactions on Quantum Computing / Physical Review A (2023) | Archival ID: QV-QC-48

1. ABSTRACT & THEORETICAL FOUNDATION

This peer-reviewed reference document synthesizes the formal mathematical formulation, operational circuit protocols, and physical implementation benchmarks for Quantum Phase Estimation (QPE) in modern quantum information architectures.

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"A super-precise speedometer for quantum rotations: it measures how fast a quantum state is spinning around a circle with microscopic precision."

3. Mathematical Formulation & Unitary Rule

$$\frac{1}{\sqrt{2^t}} \sum_{k=0}^{2^t-1} e^{2\pi i \theta k} |k\rangle \langle u| \rightarrow$$

The inverse QFT constructively focuses probability amplitudes onto the closest t-bit binary rational approximation of theta.

4. Step-by-Step Circuit Sequence

Evaluation register of t qubits + target register, Hadamard layer, cascade of controlled- U^{2^k} , QFT-dagger, measurement.

5. Executable IBM Qiskit (Python) Code

```
from qiskit import QuantumCircuit
from qiskit.circuit.library import QFT

t = 3 # 3 evaluation qubits (precision = 1/8)
qc = QuantumCircuit(t + 1, t)

# Target qubit in |1> (eigenstate of T gate with theta = 1/8)
qc.x(t)
qc.h(range(t))
```

6. Computational Complexity & Speedup

Classical Time:	Quantum Time:	Speedup Class:
$O(2^t)$ matrix multiplication or repeated gate applications	$O(t^2)$ gate complexity	Exponential

7. Key Applications & Future Research Directions

- Core engine of Shor's factoring and period finding algorithm
- Quantum simulation of molecular energies in chemistry (Quantum Chemistry QPE)
- HHL algorithm for solving linear systems

Future Work:

Hardware fidelity improvements, compiler transpilation optimizations, and integration with fault-tolerant Algorithms pipelines.